One-defect picture for long-lived optically written zigzag order in $\mathbf{Na_2Co_2TeO_6}$

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The optically written magnetic state of $Na_2Co_2TeO_6$ has two apparently conflicting signatures: the switched zigzag state lives anomalously long, implying strong pinning, while the clean microscopic landscape near the tuned $3\mathbf{Q}$–zigzag competition has only a shallow kinetic barrier and gives a sharp switching threshold. We resolve this with one extended disorder family: a structural line fault that locally enhances the Kitaev exchange. The coherent mean fault pins the written zigzag and sets the lifetime, while bond-to-bond fluctuations around the same enhanced-$|K|$ offset broaden the optical threshold. A converged climbing-image GNEB path from the pinned zigzag strip to relaxed $3\mathbf{Q}$ on a 36×36 lattice gives a quasi-two-dimensional depinning barrier $U_{\rm pin} = 21.22\,\text{meV}$, while the relaxed $3\mathbf{Q}$ endpoint remains lower by $24.15\,\text{meV}$, so the written state is metastable rather than the ground state. In driven 36×36 trajectories with the enhanced-$|K|$ line fixed and zero-mean Gaussian background disorder $\delta K_{\rm bg} \sim \mathcal{N}(0, \sigma_K)$ added to every nearest-neighbor bond, the clean pump threshold is lowered and broadened into a seed-resolved rounded magnetic melting curve. The result is a lean, experimentally testable picture in which enhanced-$|K|$ structural faults explain both the long lifetime and the rounded optical switching fraction.

I. QUESTION

The experimental phenomenology asks for one mechanism that is strong enough to pin an optically written zigzag state, but not so strong that zigzag simply becomes the equilibrium phase. In a clean near-degenerate $3\mathbf{Q}$–zigzag model, the optical response is a threshold switch: below threshold the system remains $3\mathbf{Q}$, and above threshold it writes zigzag. That clean threshold is too sharp to explain a continuous switched fraction across a disordered sample. At the same time, the measured lifetime is far too long to be explained by the clean barrier. The model therefore has to explain two different signatures:

1. a metastable written zigzag state with a large depinning barrier;
2. a disorder-broadened optical threshold, so switching appears as continuous magnetic melting rather than an all-or-nothing clean transition.

The central point of this note is that these two signatures can be obtained from one microscopic imperfection family. We use an extended enhanced-$|K|$ line as the hard lifetime fault, and we broaden the switching threshold by adding zero-mean Gaussian disorder to every nearest-neighbor bond.

II. MODEL AND OBSERVABLES

The spin Hamiltonian is the extended Kitaev model on the honeycomb lattice,

$$\mathcal{H} = \mathcal{H}_{\rm NN}(J,K,\Gamma,\Gamma') + \mathcal{H}_2(J_{2A},J_{2B}) + \mathcal{H}_3(J_3) + \mathcal{H}_7, \quad (1)$$

where $\mathcal{H}_{\rm NN}$ contains the standard nearest-neighbor J-K-Γ-Γ' exchange. The tuned Kruger parameters, in meV, are

$$\begin{aligned} J &= 0.68, & K &= -7.89, & \Gamma &= 3.07, & \Gamma' &= -2.94, \\ J_{2A} &= -0.06, & J_{2B} &= -0.70, & J_3 &= 0.52, & J_7 &= -0.40. \end{aligned} \quad (2)$$

The ring term J_7 is used only as the near-degeneracy knob: at this working point relaxed $3\mathbf{Q}$ remains slightly lower than zigzag, but the two states are close enough that an E_1 phonon pump can switch the order.

The nearest-neighbor exchange takes the standard bond-dependent form,

$$\begin{aligned} \mathcal{H}_{\rm NN} = \sum_{\langle ij\rangle_\gamma} \big[& J\,\mathbf{S}_i\cdot\mathbf{S}_j + K S_i^\gamma S_j^\gamma + \Gamma(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha) \\ & + \Gamma'(S_i^\alpha S_j^\gamma + S_i^\gamma S_j^\alpha + S_i^\beta S_j^\gamma + S_i^\gamma S_j^\beta)\big], \end{aligned} \quad (3)$$

where $\{\alpha,\beta,\gamma\}$ is a cyclic permutation of $\{x,y,z\}$ for each bond type (so $\alpha = y, \beta = z$ on an x-bond, etc.) and spins are in the local Kitaev frame.

The driven dynamics couples the spins to the zone-center E_1 infrared-active optical phonon of NCTO. The compound has space group $P6_322$ (point group D_6); the IR-active doublet is the two-component normal coordinate $\mathbf{Q} = (Q_x, Q_y)$, the in-plane relative displacement of the two Co sublattices, with conjugate momentum $\mathbf{P} = (P_x, P_y)$. We treat it as a single zone-center mode (uniform across the lattice) with unit effective mass. The complete Hamiltonian is

$$\mathcal{H}_{\rm total} = \mathcal{H}_{\rm spin} + \mathcal{H}_{\rm ph} + \mathcal{H}_{\rm ME} + \mathcal{H}_{\rm drive}. \quad (4)$$

The phonon is a single two-dimensional harmonic oscillator in $\mathbf{Q}$,

$$\mathcal{H}_{\rm ph} = \tfrac{1}{2}\,|\mathbf{P}|^2 + \tfrac{1}{2}\,\omega_{E_1}^2\,|\mathbf{Q}|^2, \quad (5)$$

optionally with an anharmonic term $\frac{1}{4}\lambda_4|\mathbf{Q}|^4$ (set to zero here). The magnetoelastic coupling is the leading magnetic response to this displacement: each nearest-neighbour exchange constant X acquires a $\mathbf{Q}$-dependent shift on a γ-bond,

$$\delta X_\gamma(\mathbf{Q}) = \lambda_{X,0}\,|\mathbf{Q}|^2 + \lambda_{X,2}\big[(Q_x^2 - Q_y^2)\cos 2\theta_\gamma + 2Q_xQ_y \sin 2\theta_\gamma\big], \tag{6}$$

for $X \in \{J, K, \Gamma, \Gamma'\}$, with bond-axis angles $\theta_x = 0$, $\theta_y = 2\pi/3$, $\theta_z = 4\pi/3$. The shift is *quadratic* in $\mathbf{Q}$ because a term linear in $\mathbf{Q}$ cancels once the three bond types are summed and is therefore symmetry-forbidden; Eq. (6) is the leading C_6-invariant form. Here $\lambda_{X,0}$ is an isotropic (breathing) deformation potential and $\lambda_{X,2}$ the anisotropic (E_1 form-factor) deformation potential; we set $\lambda_{X,0} = 0$ throughout. Since δX_γ sits exactly where the bare constant X appears in Eq. (3), the magnetoelastic Hamiltonian has the same bilinear form as $\mathcal{H}_{\rm NN}$ with every coupling replaced by its displacement-induced shift,

$$\begin{aligned}\mathcal{H}_{\rm ME} = \sum_{\langle ij\rangle_\gamma} \big[&\delta J_\gamma(\mathbf{Q})\,\mathbf{S}_i\cdot\mathbf{S}_j + \delta K_\gamma(\mathbf{Q})\,S_i^\gamma S_j^\gamma \\ &+ \delta\Gamma_\gamma(\mathbf{Q})\,(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha) \\ &+ \delta\Gamma'_\gamma(\mathbf{Q})\,(S_i^\alpha S_j^\gamma + S_i^\gamma S_j^\alpha + S_i^\beta S_j^\gamma + S_i^\gamma S_j^\beta)\big],\end{aligned} \tag{7}$$

where each shift $\delta X_\gamma(\mathbf{Q})$ is given by Eq. (6) and $\{\alpha, \beta, \gamma\}$ is the same cyclic permutation as in Eq. (3). A resonant Gaussian THz pulse couples polarly to the mode through its effective charge Z^*,

$$\mathcal{H}_{\rm drive} = -Z^*\,\mathbf{E}(t)\cdot\mathbf{Q}, \tag{8}$$

with $\mathbf{E}(t) = E_0\,\hat{\mathbf{e}}\,\exp[-(t-t_0)^2/2\sigma_t^2]\cos(\omega_{E_1}t)$: pump amplitude E_0, polarization $\hat{\mathbf{e}}$, center t_0, and width σ_t. Phonon dissipation enters the equations of motion as a damping rate γ_{E_1}, and the spin dynamics carries Gilbert damping α.

Driven trajectories adopt the isotropic Grüneisen choice in which every channel shares a common fractional modulation $\delta X/X$; for the quadratic striction model this means the anisotropic deformation potential scales linearly *and with sign* with the bare exchange constant, $\lambda_{X,2} = (X/K)\,\lambda_{K,2}$. Fixing $\lambda_{K,2} = 0.02$ gives $\lambda_{J,2} = -0.0017$, $\lambda_{\Gamma,2} = -0.0078$, and $\lambda_{\Gamma',2} = +0.0075$. The phonon and pulse parameters are $\omega_{E_1} = 4.0\,$meV, $\gamma_{E_1} = 0.0849$, $Z^* = 1$, pump center $t_0 = 50$, width $\sigma_t = 15$, Gilbert damping $\alpha = 0.05$, RK4 time step 0.005. All production results emphasized below use $L = 36$ honeycomb lattices ($N = 2592$ spins): the selected hard-fault GNEB path, the fixed-line global-disorder switching fraction, and the regenerated defect geometry are all evaluated on the same solver lattice convention. With Grüneisen-scaled all-channel coupling the clean fixed-line threshold lies between $E_0 = 20$ and 22 (the seed-0 trajectory is off at $E_0 = 20$ and fully switched at $E_0 = 22$); global disorder lowers the 25% crossing from $E_0 \simeq 21$ down to $E_0 \simeq 10$–14 as σ_K grows. The order diagnostic

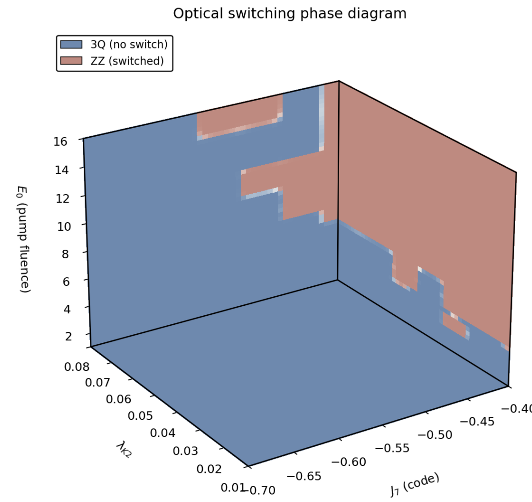

FIG. 1. Clean driven phase diagram at the tuned-Kruger point. The clean system is the control problem: it has a comparatively sharp threshold between a stable $3\mathbf{Q}$ response and a zigzag-like written state. This cube fixes the microscopic switching physics that the large-cell disorder calculations broaden.

is the ratio

$$r_3 = \frac{\min_i S(\boldsymbol{M}_i)}{\max_i S(\boldsymbol{M}_i)}, \tag{9}$$

over the three zigzag wave vectors. We also use a local zigzag overlap to count the fraction of sites written into the targeted zigzag domain after a quench.

For kinetic barriers we use the climbing-image geodesic nudged-elastic-band method on the curved spin manifold. The GNEB path shown below is the physical depinning path from the pinned zigzag strip to relaxed $3\mathbf{Q}$ with the defect present. The energy zero is the pinned zigzag endpoint; the right endpoint is the relaxed $3\mathbf{Q}$ state. Thus a negative final endpoint in the plot means that $3\mathbf{Q}$ is lower in energy and the written zigzag state is metastable.

III. CLEAN SWITCHING LANDSCAPE

Figure 1 is the clean baseline. It shows why disorder must be included at all. The pump can access a zigzag-like basin near the tuned $3\mathbf{Q}$–zigzag competition, but in a clean cell this response is organized by a threshold. The experimental switching fraction is smoother than this clean limit, so the disorder calculation should be judged by whether it broadens the threshold without making zigzag the equilibrium phase before the pump arrives.

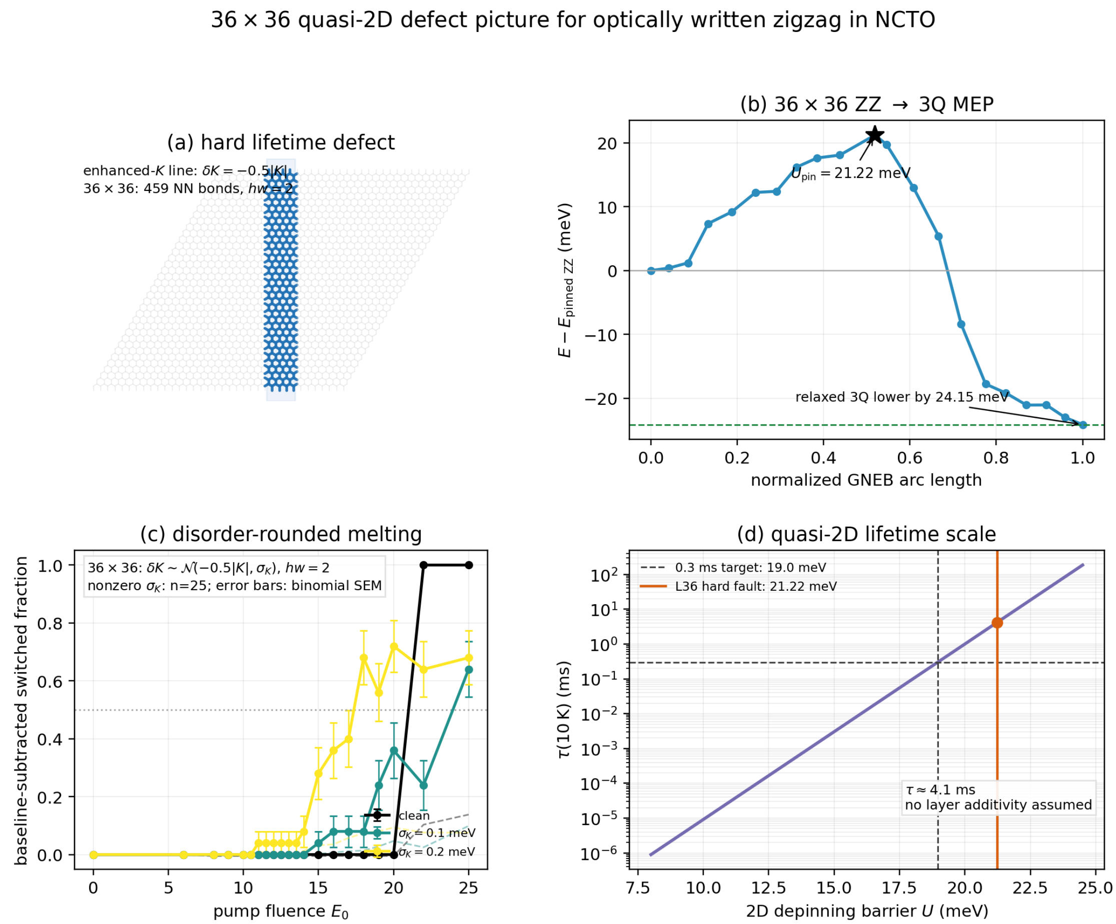

FIG. 2. One-defect account of the experimental signatures. **(a)** The hard lifetime defect is an extended line/band of nearest-neighbor bonds on which the Kitaev magnitude is enhanced: $\delta K = -0.5|K|$, i.e. K is driven more negative without a sign change. The selected geometry is a vertical band of half-width $hw = 2$ on the 36×36 honeycomb and contains 459 perturbed bonds. **(b)** Large-cell GNEB minimum-energy path from pinned zigzag to relaxed $3\mathbf{Q}$ with this defect present. The saddle is image 10 and gives $U_{\text{pin}} = 21.22\,\text{meV}$ above the pinned zigzag endpoint; the relaxed $3\mathbf{Q}$ endpoint lies $24.15\,\text{meV}$ lower, so the pinned zigzag is metastable, not the ground state. **(c)** The continuous switching/melting signature is captured on the same 36×36 lattice by the same enhanced-$|K|$ line held fixed at $\delta K_{\text{line}} = -0.5|K|$, with zero-mean Gaussian background K disorder applied to all nearest-neighbor bonds. Points are baseline-subtracted switching fractions with binomial standard-error bars; dashed curves are excess quenched zigzag fractions. **(d)** The quasi-two-dimensional Arrhenius estimate uses the 36×36 in-plane depinning barrier directly.

IV. ONE PICTURE

Figure 2 is the whole proposed mechanism. The lifetime pin is not a nematic field and not a smooth random potential. It is a hard structural fault: a vertical line/band of nearest-neighbor bonds where the Kitaev exchange is locally enhanced, $\delta K = -0.5|K|$. Because $K < 0$, this makes K more negative and increases $|K|$; it does not flip the sign of K. The selected band has half-width $hw = 2$ and covers 459 nearest-neighbor bonds in the 36×36 calculation. This is the best switchable point in the extended-defect catalogue: it has the largest converged depinning barrier among defects for which relaxed $3\mathbf{Q}$ remains the lower-energy state.

The GNEB panel makes the metastability explicit. The pinned zigzag endpoint is the left endpoint and defines zero energy. The path climbs to $21.22\,\text{meV}$ and then descends to the relaxed $3\mathbf{Q}$ endpoint, which is $24.15\,\text{meV}$ below pinned zigzag. Therefore the defect does not make zigzag the equilibrium phase. It creates a local trap: zigzag can be written and pinned, but the surrounding equilibrium order is still $3\mathbf{Q}$. This is exactly the "switchable" regime. The large-cell path ran for 1600 iterations with a stable saddle image and a last-window barrier scatter of $0.18\,\text{meV}$, already safely above the $18.96\,\text{meV}$ lifetime target.

The switching/melting panel uses the same defect family. The selected line is kept at the lifetime value $\delta K_{\text{line}} =$

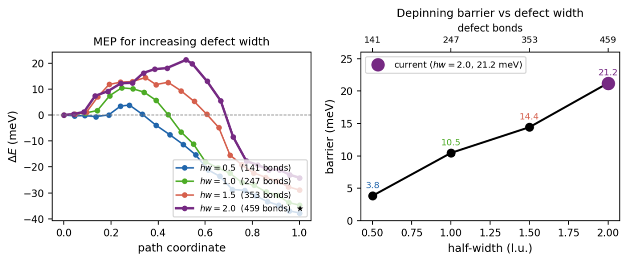

FIG. 3. Kinetic barrier of the hard-fault defect as a function of band width. (Left) Climbing-image geodesic MEP curves for four defect half-widths ($hw = 0.5$, 1.0, 1.5, 2.0; 141, 247, 353, 459 defect bonds) at fixed $\delta K = -0.5|K|$. The path runs from the pinned zigzag strip to relaxed $3\mathbf{Q}$ with the defect present; energies are shifted so the zigzag endpoint lies at zero. (Right) Converged barrier height versus half-width. The barrier grows monotonically from 3.8 meV at $hw = 0.5$ to 21.2 meV at the operating point $hw = 2$ (star), which is safely above the 18.96 meV lifetime target. The relaxed $3\mathbf{Q}$ endpoint sits 24.15 meV below the pinned strip at the operating point, confirming metastability.

$-0.5|K|$. To broaden the optical switching fraction we do not vary the strength of that line; instead we add the earlier zero-mean Gaussian background disorder to every nearest-neighbor bond, $\delta K_{\rm bg} \sim \mathcal{N}(0, \sigma_K)$. This separates the two roles: the extended line fixes the hard pinning scale, while the bulk background inhomogeneity distributes local pump thresholds. With all-channel coupling the clean fixed-line threshold lies between $E_0 = 20$ and 22. Global disorder lowers and broadens the onset: in the twenty-five-seed scan, $\sigma_K = 0.2$ meV first shows appreciable switching above $E_0 \simeq 14$ and reaches $\sim 70\%$ by $E_0 = 18$–20; $\sigma_K = 0.3$ meV reaches 88% at $E_0 = 25$ with onset beginning near $E_0 \simeq 14$; $\sigma_K = 0.5$ meV shows switching from $E_0 = 6$ onward, saturating near 60–64%. The normalized-onset 25–75% widths are 1.00, 5.30, 3.00, 6.79, 4.19, and 5.00 in E_0 for $\sigma_K = 0$, 0.1, 0.2, 0.3, 0.4, and 0.5 meV, respectively. This is the cleanest switching-fraction cross-check: the line defect is held fixed, and only the background disorder controls the rounding.

V. LIFETIME SCALE

The Arrhenius estimate is

$$\tau = \tau_0 \exp\left(\frac{U}{k_B T}\right), \qquad \tau_0 = \frac{\hbar}{|K|} = 0.0834\,\text{ps}. \tag{10}$$

At $T = 10$ K, a 0.3 ms lifetime requires

$$U_{0.3\,\text{ms}} = k_B T \log\left(\frac{0.3\,\text{ms}}{\tau_0}\right) = 18.96\,\text{meV}. \tag{11}$$

The 36×36 quasi-two-dimensional hard-fault barrier is 21.22 meV. With the same attempt time this gives $\tau \simeq 4.1$ ms at 10 K, above the 0.3 ms target. Since NCTO is treated here as a quasi-two-dimensional magnet, this in-plane barrier is the lifetime estimate used in the paper.

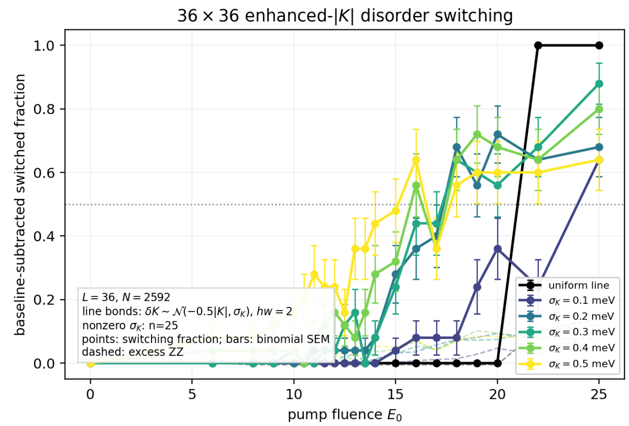

FIG. 4. Large-cell switching-fraction check on a 36×36 honeycomb lattice ($N = 2592$ spins). The calculation uses the same enhanced-$|K|$ line as the lifetime MEP: selected bonds in a $hw = 2$ vertical band have the fixed offset $\delta K_{\rm line} = -0.5|K|$, while all nearest-neighbor bonds also receive zero-mean background disorder $\delta K_{\rm bg} \sim \mathcal{N}(0, \sigma_K)$. All geometry, reference states, disorder files, driven MD runs, and quenches are regenerated on the 36×36 solver lattice. The displayed pump range covers the broadened onset of the fixed-line/global-disorder protocol. Nonzero-disorder curves use twenty-five seeds; error bars in Fig. 2(c) denote the binomial standard error of the switching fraction.

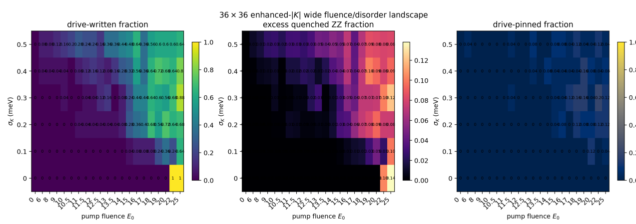

FIG. 5. Fluence/disorder map for the fixed-line/global-background protocol on the 36×36 lattice. The enhanced-$|K|$ line is held fixed at $\delta K_{\rm line} = -0.5|K|$ for every row, while the background Gaussian width σ_K varies across all nearest-neighbor bonds. The left panel is the baseline-subtracted drive-written fraction, the middle panel is the excess quenched zigzag fraction, and the right panel is the pinned-written fraction.

VI. OTHER SCENARIOS CONSIDERED

The one-picture mechanism is the result of ruling out several cleaner-looking but less robust explanations.

a. Clean intrinsic barrier. The clean near-degenerate model switches sharply and does not supply a robust millisecond barrier. The clean GNEB landscape is too shallow and too spatially uniform; it can explain a threshold but not a long pinned lifetime.

b. Classical nucleation theory. The CNT droplet language was useful for scale estimates, but it is not the right story for the observed metastability. A finite zigzag patch in a clean $3\mathbf{Q}$ background is not guaranteed to be a stable droplet. It either shrinks or grows depending on the local bias and wall motion. The long lifetime has to come from boundary depinning at real defects, not from a clean bulk nucleation barrier.

c. Point defects and vacancies. Vacancies or isolated bond defects perturb too few bonds. They can locally distort a domain wall, but their depinning barriers remain well below the 19 meV target unless the perturbation becomes so large and extended that it is no longer a point defect. Bond-vacancy scans also do not give a clean monotonic route to a switchable high barrier.

d. Kitaev sign flip. A local K sign flip is a strong microscopic perturbation, but it is not the best physical candidate. In the catalogue it does not outperform the enhanced- $|K|$ fault in the switchable regime, and it is harder to motivate as a modest structural distortion of the Co–O–Co exchange path.

e. Nematic fault as the lifetime pin. Nematic defects are efficient at selecting zigzag, but this is exactly why they are dangerous. A strong nematic bias couples directly to the global zigzag director and tends to make zigzag the ground state. That is over-pinning, not a reversible metastable written state. We therefore do not use nematic as the final switching or lifetime defect.

f. Global zero-mean K disorder. The final switching calculation keeps the enhanced-$|K|$ line fixed at $\delta K_{\mathrm{line}} = -0.5|K|$ and adds zero-mean Gaussian disorder $\delta K_{\mathrm{bg}} \sim \mathcal{N}(0, \sigma_K)$ to every nearest-neighbor bond. This separates the two roles cleanly: the fixed line controls the hard depinning scale, while the global bond inhomogeneity distributes local pump thresholds without introducing a second defect species.

g. Instantly adiabatic strain along GNEB. We do not assume that strain dynamically follows every image during a real pump trajectory. The GNEB calculation is a static transition-state calculation for the spin barrier with the chosen defect present. The switching simulations are separate driven dynamical calculations. This distinction is important: the MEP establishes the thermal depinning barrier of the written state; it is not meant to describe the full nonequilibrium phonon-assisted writing process image by image.

VII. EXPERIMENTAL SIGNATURES

This picture makes four directly checkable claims. First, samples with more extended in-plane structural faults should show longer written-state lifetimes, with an exponential sensitivity to the two-dimensional depinning barrier. Second, the written state should remain metastable: after removing the pump, relaxed 3**Q** is still the lower-energy state away from the defect. Third, the optical switching fraction should broaden continuously as the enhanced-$|K|$ line becomes spatially inhomogeneous, not jump as a perfectly clean threshold. In the present 36×36 statistics this appears as fractional switching at the onset fluences: with $\sigma_K = 0.3$ meV the drive-written fraction rises from 1/4 around $E_0 = 8$–9 to 3/4 by $E_0 = 10.5$. Fourth, the lifetime and the switching width probe different aspects of the same fault family: the mean offset controls the depinning barrier, while the width of the bond-offset distribution controls the threshold distribution.

The resulting story is deliberately narrow. A hard enhanced-K structural line fault pins the boundary and gives a converged ZZ-to-3Q depinning barrier. The The same line, with global zero-mean Gaussian disorder added to all nearest-neighbor bonds, broadens the optical threshold into a fractional seed-resolved onset. Together they reconcile the sharp lifetime signature with continuous magnetic melting without making zigzag the equilibrium state.

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